CS2009- DESIGN AND ANALYSIS OF ALGORITHMS

Assignment 1 — Fall 2026

Total Marks	Due Date	Late Policy
100	**19 September 2026**	**20% 1-day late, 40% 2-day late** **No Submissions afterwards**

Instructions

- Show all important steps of your solution.
- Zero marks on plagiarism.
- Scanned copy of the handwritten assignment in PDF format is required to submit.

Question 1 — Insertion Sort [8 Marks]

Consider the following list of integers: **18, 7, 25, 14, 3, 21, 10, 30**

1. Show the steps performed by Insertion Sort to arrange the list in ascending order.
2. At each iteration, clearly show the current array and the value of the key variable.
3. Explain why the worst-case running time of Insertion Sort is $O(n^2)$.
4. Use a loop invariant to explain why Insertion Sort correctly sorts the array.

Question 2 — Merge Sort [8 Marks]

Consider the following list**: 32, 15, 8, 24, 11, 40, 6, 19**

1. Show the division, recursive sorting, and merging steps of Merge Sort to arrange the list in ascending order.
2. Suppose a variation of Merge Sort divides the input into three parts at each recursive level, recursively sorts the three parts, and then combines them using a three-way merge. Determine the overall asymptotic running time and briefly justify your answer.

Question 3 — Quick Sort [8 Marks]

Consider the following list: **24, 9, 17, 5, 31, 12, 27, 3**

1. Apply Quick Sort using the first element as the pivot. Show the array after each major partition step.
2. Clearly identify the pivot used at every partition step.
3. Use a loop invariant to explain why the partition procedure correctly places the pivot.

Question 4 — Dividing and Conquer [8 Marks]

Suppose you have 2n players, and each player has a numerical skill rating. You need to divide them into two teams of n players each so that the teams are as balanced as possible in terms of their total skill ratings.

For example, consider the ratings: **10, 15, 7, 20, 12, 18, 9, 14**

1. Design a simple $O(n^2)$ algorithm for dividing the players into two reasonably balanced teams. Explain the basic idea.
2. Design an O (n log n) approach using sorting. Explain the strategy and analyze its running time.

Question 5 — Pairing Numbers [8 Marks]

You are given 2n real numbers. Divide them into n pairs so that the largest sum among all pairs is as small as possible. For the following input: **2, 4, 6, 8, 10, 12**

1. Find a pairing that minimizes the largest pair sum. Show all pair sums and identify the maximum pair sum.
2. Design an O (n log n) algorithm for solving this problem. Explain why your strategy gives an optimal pairing.

Question 6 — Asymptotic Proofs [8 Marks]

(a) Prove that: $n^2 + 5n + 10 = O(n^2)$

Determine suitable values of c and n_o.

(b) Prove that: $4n^2 \log_2 n + 3n^2 = O(n^2 \log_2 n)$

Determine suitable values of c and n_o. Show all important steps.

Question 7 — Asymptotic Notations [8 Marks]

Watch this video lecture on Big-O, Big-Ω and Big-Θ notation. Answer the following in your own words.

1. What is the main purpose of asymptotic notation in algorithm analysis? [2 Marks]
2. Explain the difference between Big-O, Big-Ω, and Big-Θ. [2 Marks]
3. Why are constants and lower-order terms generally ignored when analyzing asymptotic running time? [1 Mark]
4. Give one simple example of two algorithms whose growth rates are different as the input size increases. [1 Mark]
5. Write a short summary of the video lecture and mention at least two important concepts you learned. [2 Marks]

Note: Copying the lecture transcript or material from another source will not receive full credit.

Question 8 — Master Theorem [12 Marks]

Use the Master Theorem to determine the asymptotic running time of each recurrence. For each recurrence, identify a, b, and f(n), and state the appropriate case.

1. $T(n) = \sqrt{2}\, T(n/2) + \log n$
2. $T(n) = 64\, T(n/8) - n^2 \log n$
3. $T(n) = 16\, T(n/4) + n!$

Question 9 — Iteration Method [8 Marks]

Use the Iteration Method to determine the asymptotic running time of each recurrence. Show the expansion for several levels, identify the pattern, and state the stopping condition.

(1) $T(n) = 2T(n/2) + n\log n$

(2) $T(n) = 8T(n/2) + n^2$

Question 10 — Asymptotic Growth [10 Marks]

For each statement, write True (T) or False (F) and justify your answer using an example, mathematical reasoning, or a suitable counterexample.

(a) For every positive function f(n), $f(n) + O(f(n)) = \Theta(f(n))$.

(b) If $f(n) = O(g(n))$ and $g(n) = O(h(n))$, then $f(n) = O(h(n))$.

(c) If $f(n) = \Theta(g(n))$, then $f(n) + g(n) = \Theta(g(n))$.

(d) $n^2 = O(n^3)$.

(e) $n^3 = O(n^2)$.

Question 11 — Recursion Tree Method [7 Marks]

(a) A distributed log-processing system divides an input of size n into three independent batches of size n/4. Coordinating and merging the results requires n log n additional operations.

$$T(n) = 3T(n/4) + n \log n, \quad T(1) = \Theta(1)$$

Draw the recursion tree, derive the cost at level i, determine the tree height, identify the dominant level, and obtain a tight asymptotic bound. [4 Marks]

(b) An image-analysis algorithm creates four subproblems of size n/2 and performs n^2 work to combine their results.

$$T(n) = 4T(n/2) + n^2, \quad T(1) = \Theta(1)$$

Use a recursion tree to find the cost of each level and the number of levels. Explain why no single internal level dominates the total cost, and state the tight asymptotic bound. [3 Marks]

Question 12 — Substitution (Guess-and-Test) Method [7 Marks]

(a) A parallel report generator divides a report into two equal sections. Combining the section summaries requires n log n operations.

$$T(n) = 2T(n/2) + n \log n, \quad T(1) = \Theta(1)$$

Guess a tight asymptotic bound and prove the upper bound using substitution and mathematical induction. Clearly state the induction hypothesis, substitute it into the recurrence, and specify suitable constants and a base case. [4 Marks]

(b) A compression routine recursively processes two blocks of size n/4 and performs $\sqrt{n}$ preprocessing operations.

$$T(n) = 2T(n/4) + \sqrt{n}, \quad T(1) = \Theta(1)$$

Guess a tight asymptotic bound and verify the upper bound using the Substitution Method. Your proof must show the inductive step and explain why the guessed logarithmic factor is necessary. [3 Marks]

End of Assignment